



Public and Private Sector Earnings of Immigrants and the Canadian-Born: Evidence from the Labour Force Survey

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Abstract

Using Statistics Canada's monthly Labour Force Survey master files from January 2006 to December 2018, this paper evaluates the wage differences between immigrants and comparable Canadian-born workers both within the respective levels of the private and public sectors, as well as the public sector wage premium within the immigrant and Canadian-born populations. Using both Ordinary Least Squares and Oaxaca-Blinder decomposition techniques, we find that the immigrant wage gap tends to be negative and is largest in the private sector. The public-sector premium relative to the private sector is also larger for immigrants than for the Canadian-born when we compare wage differences within the two groups. Combined, these results suggest that public sector wages are important to narrowing the overall immigrant wage gap.

Keywords Canada · Immigrants · Private sector · Public sector · Wage differentials

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Introduction

Nearly a quarter of Canada's total population are immigrants and this proportion will keep increasing (Statistics Canada, 2017). One challenge arising from the growing immigrant population in Canada is ensuring wage equity between the Canadian-born and immigrant workers across all sectors of the Canadian labour market. With Canada's aging population and the growing significance of immigrant workers in the Canadian labour market, "Canada's global competitiveness in the future will be in part determined by the degree that immigrants will be able to integrate into the Canadian labour market" (King, 2009:2). As Kerr and Kerr (2011) highlight in their study, assessment of successful integration of immigrants into a host-country's labour market can primarily be ascertained through a comparison of wages (and employment rates) of immigrants versus the host-country's workers.

Overall, existing evidence reveals that there are significant wage differences between the Canadian-born and immigrants in the Canadian labour market (Banerjee & Lee, 2015; Frenette & Morissette, 2005; Picot & Hou, 2014; Green & Worswick, 2010; Nadeau, 2013). Immigrants in Canada on average earn significantly less than their Canadian-born counterparts, although the former group may have equal or higher qualifications than the latter group (Hou & Coulombe, 2010; Nadeau, 2013). These are captured in the literature as the immigrant wage gap, defined as the difference between immigrant wages and those of the Canadian-born with similar characteristics (Zheng, 2017).

To date, however, the literature is largely silent on if and how immigrant wages compare to the Canadian-born in both the private and public sectors. It is the aim of this research to fill this gap. In particular, this research documents the differences in wages between immigrants and the Canadian-born in both the private and public sectors, and also estimates any public sector wage premium within these two demographic groups. This exercise is important and will assist policymakers in ascertaining where wage differentials exist with the ultimate goal of the better utilisation of immigrant human capital in the Canadian labour market.

Literature Review

Factors such as education, geographic settlement area and immigrant source countries are consistently shown in the literature to contribute to the overall immigrant wage gap. For instance, Green and Worswick (2010) find that the wage gap is wider among immigrants with foreign education than among those who acquired their education in Canada. In analysing the performance of university-educated immigrants in the Canadian labour market by geographic settlement areas, King (2009) showed a negative relationship between immigrant population density and immigrant earnings. Using the 2006 census, King found that Ontario, which attracts the highest number of immigrants, had the highest immigrant wage gap of 29.2%, while this relationship did not exist for immigrants who settled in sparsely populated immigrant provinces like Newfoundland and Labrador (where in fact immigrants on the average earn more than their Canadian-born counterparts).

The little empirical evidence that does exist to date on the topic of immigrant wage differentials between the private and public sectors suggests that the observed immigrant gap is more prevalent in the private sector than in the public sector (Nadeau, 2013; Zheng, 2017). Coulombe et al. (2014) find that Canadian employers generally assess the quality of human capital to be lower if credentials are obtained outside of Canada. But Nadeau (2013) maintains that government employers may be in a better position to accurately evaluate foreign credentials because they are relatively large and so may have superior resources to translate these accurately into their Canadian equivalents. Also, factors such as unions' abilities to hold government accountable to its pay equity legislation (Zheng, 2017) and pro-immigration governments' vested interest in ensuring the success of immigrants in the labour market may account for this narrower immigrant wage gap.

A consistent thread among both the immigrant and the Canadian-born groups is that public sector workers record a wage premium over their respective private sector counterparts (Zheng, 2017), and this reflects the finding of the broader literature on public-private sector wage differentials in Canada (Mueller, 1998, 2000; Lammam et al., 2016; Palacios & Clemens, 2013; Prescott & Wandschneider, 1999; Shapiro & Stelcner, 1989; Tiagi, 2010).

Some other studies have proposed that the promotional rules in the public sector, unlike the private sector, reward quantity (e.g. seniority, years of schooling and experience) over quality (McIvor, 2016; Nadeau, 2013). The most prominent among these reasons is the theoretically sound argument that the public sector (unlike the private sector) is not constrained by the profit-maximisation condition, but rather often operates under monopolistic conditions and thus able to reward labour with higher wages (Mueller, 2000; Nadeau, 2013; Lammam et al., 2016). Political re-election considerations also play a vital role in determining wages in the public sector (Zheng, 2017). Gunderson (1979:230) succinctly summarises this saying that "the basic difference between the public and private sector is that in the former the profit constraint is replaced by an ultimate political constraint." In a similar vein, the public sector often has structured pay scales which may leave less room for discretion in wage offerings compared to the private sector.

Furthermore, the wage gap between the sectors is partially attributable to differences in worker endowments and (arguably) that the public sector could employ the "cream of crop" and reward workers accordingly (Tiagi, 2010). Indeed, McIvor (2016) too finds that public sector workers are on the average more educated than workers in the private sector and, consistent with human capital theory, would have been expected to have higher wages.

Even within the public sector, wage premium rates vary across respective levels of government. The premium is highest for federal government workers than for those at the provincial and local government levels (Mueller, 2000). Consistent with Mueller's earlier finding, Mallett and Wong (2008) also conclude that the public sector wage premium (relative to the private sector) is highest at the federal level (17.3%), and second highest at the municipal level (11.2%) while the provincial level (7.9%) records the lowest premium.

While all public sector workers have been shown to earn wage premiums in the public sector, the overall impact of this benefit may be mitigated for immigrants owing to their relatively small numbers in that sector. Recently, the proportion of immigrants

in the public sector (13% for males and 27.5% for females) has been shown to be significantly lower than that of the Canadian-born (19.6% for males and 36.6% for females) (Nadeau, 2013). This could be because Canadian citizenship is a requirement for employment in most public sector jobs, especially in federal public administration (Zheng, 2017). Furthermore, several occupations in the public sector such as those in health care tend to be regulated (i.e. barriers to entry), making it more difficult for skilled immigrants to enter (Zheng, 2017). On the other hand, the Public Service Employment Act, a legislative instrument for hiring priorities for public sector workers, is seen to apportion (minimum) hiring quotas for females, immigrants and other minority groups (McIvor, 2016).

While the overall representation of immigrants in the public sector may be beneath their overall population share, there are signs of improvement. For example, the federal public service appears to be increasing its representation of immigrants. While immigrants are not addressed as a separate equity seeking group by the federal government, visible minorities are counted, and immigrant status and visible minority status are highly and positively correlated. The Treasury Board of Canada Secretariat (2020), for example outlines two measures to determine the representation of four designated employment equity groups (i.e. women, Indigenous peoples, persons with disabilities and members of visible minorities). The first of these measures is labour market availability (LMA) which shows the population share of each of these four groups. By this measure, the core federal public administration in 2018–2019 was comprised of 16.7% visible minority members, less than their 21.7% share of the Canadian population. The second definition is a subset of LMA, work force availability (WFA), and applies additional criteria such as education levels and citizenship. By this measure, 15.3% of the core federal public administration were visible minorities in 2016, up from 13.0% in 2011. This latter measure is likely the most important for the purposes of comparing immigrants to the Canadian-born in the public sector since the former group may be lacking Canadian citizenship and relevant education.

The same report shows that had high representation in major (public sector) occupations such as financial management (30.5%), auditing (26.8%) architecture, engineering and land survey (22.1%), applied science and patent examination (21.9%), information technology (21.5%) and economics and social science services (21.1%). They however had low representation in operational services (4.7%) and health services (11.1%). Related data show that the federally regulated private sector also increased visible minority representation between 2016 and 2017, up from 22.2 to 22.8% (Employment and Social Development Canada, 2019). Still, the Treasury Board Secretariat (2020:3) notes that “... additional progress could be made in terms of members of visible minorities reflecting their LMA.” Whether or not there also might be improvements in terms of the labour market earnings of immigrants in the public sector is the topic of much of what follows.

To the best of our knowledge, there have only been two studies to date on the subject of immigrant-Canadian-born wage comparison in the public and private sectors. Nadeau (2013) finds male and female immigrants, respectively, earn 3.1% and 0.6% more than their Canadian-born counterparts in the public sector, compared to in the private sector where they earn 3.5% and 0.5% less. Zheng (2017), on the other hand, finds no immigrant wage gap between male immigrants and their Canadian-born counterparts in the public sector but estimates an immigrant wage gap of 4.1% in the

private sector. For females, however, he finds an immigrant wage gap of 2.9% in the public sector and 2.7% in the private sector.

Applying decomposition techniques, Nadeau (2013) finds that the unexplained effect component is the largest portion of the immigrant wage gap within the private sector. He goes on to tease out the effect of discrimination from this unexplained portion using the immigrants' country of birth as a proxy. He estimates that discrimination is more prevalent in the private sector than in the public sector. In fact, he finds that there is no discrimination among females in the public sector. For the private sector, however, he finds that male immigrants from non-traditional source countries (i.e. Africa, Asia) earn 18.8% less than their Canadian-born counterparts, whereas immigrants from traditional source countries (i.e. USA, Western Europe) earn wages as much as 2.8% higher than comparable Canadian-born individuals.

Zheng (2017) also decomposes public-private sector wage differentials among immigrants and the Canadian-born finds that the unexplained effect accounts for the larger proportion of the total public-private sector wage gap among immigrants. He is unable to draw a similar conclusion for the Canadian-born.

These studies, however, do not evaluate the wage differentials between immigrants and the Canadian-born at the respective levels of the public sector. These levels include employees not involved in public administration (most who are in health care and education) and those involved in public administration at the federal, provincial and local levels of government. It is uncertain whether the relatively smaller immigrant wage gap observed in the public sector cuts across all levels of the public sector, or if it is just one or two levels in the sector that could be accounting for these observed differences. More so, these two studies on the subject do not (and indeed often cannot given the data used) control for certain key job, industry and occupational characteristics such as union status of workers, all of which may have confounding effects on any wage premiums.

This study therefore seeks to fill in these gaps. Specifically, we seek to address the following research questions: Is the immigrant wage gap smaller at different levels of the public sector relative to the private sector? And is the premium (or penalty) for immigrants in the public and private sectors different from that of the Canadian-born?

Data and Summary Statistics

Statistics Canada's Labour Force Survey (LFS) master file cross-sectional data from January 2006 through December 2018 (i.e. 300 months) are utilised in the empirical work that follows. Merging these monthly datasets gives a larger sample size for the analysis. This period not only covers all periods under which Nadeau (2013) and Zheng (2017) have analysed, but also more recent years. And, unlike Nadeau (who uses the 2006 Census), and Zheng (who uses the 2011 National Household Survey), the LFS contains more detailed information on several important variables (e.g. union status).

This work identifies the Canadian-born group and the immigrant group respectively from the variables "CNTRYBTH" (country of birth) and "LANDIMM" (landed immigrant). The Canadian-born group includes those who recorded their country of birth to be Canada. The variable "LANDIMM" has a universe of all those who indicate their country of birth not to be Canada and further it indicates whether they are landed

immigrants or not. The largest number of the “LANDIMM” respondents are landed immigrants (i.e. permanent residents and citizens), whereas a small number consist of Canadians-by-birth who are born to Canadian parents outside of Canada as well as temporary residents (i.e. those born abroad who are neither landed immigrants nor citizens). These groups are eliminated from the immigrant group to make the estimates clearer without the confounding influences of Canadians born abroad to Canadian parents and temporary residents. As Zheng (2017) highlights, a significant proportion of temporary residents could return to their home country at the expiration of their visas. Also, most public sector jobs require citizenship or permanent residency status. Hence, the final definition of the immigrant group is people born outside of Canada and who are landed immigrants (i.e. permanent residents or citizens but not those born as citizens outside of Canada).

Private and public sector employees are identified by the variable “COWMAIN” (class of worker at main job). This variable also records different categories of self-employed workers, which are eliminated because there are no wage data for that group of workers. The “NAICS” (North American Industry Classification System) variable is used to further identify the levels of the public sector: federal government public administration; provincial and territorial public administration; local, municipal and regional public administration; Aboriginal public administration (a very small sample fraction 0.13%); international and other extra-territorial public administration (also a very small sample fraction of 0.19%) and the non-administration public sector (i.e. all other NAICS categories owned or funded by the government). The analysis that follows focuses on the largest subgroups within the public sector: federal public administration, provincial public administration, local public administration and non-administration public sector.

A sample restriction on the age of workers is imposed so that the sample includes only those between 25 and 54 years, common in the literature to capture the most active labour force. Those who work less than 5 h a week or more than 100 h a week are also eliminated, as are those with real hourly wages (in 2002 dollars) of less than \$5. The final sample contains 5,333,078 observations: 815,435 immigrants and 4,157,643 Canadian-born individuals.

Selected summary statistics for immigrants and Canadian-born workers employed in both the public and private sectors are presented in Table 1.¹ Both immigrant and Canadian-born public sector workers generally have higher average real wages than their respective private sector counterparts. Within the sectors however, the average real wage differential is highest in the private sector for the Canadian-born where the average real hourly wage is \$25.25, or 9.6% higher than for immigrants. Also, the Canadian-born record higher average real wages overall in the public sector but this difference is not as great: \$24.67 versus \$23.80, or about 3.1% higher. These higher wages for the Canadian-born, however, differ by subsector with the largest difference in the non-administration public sector (3.7%) and the smallest difference—in fact a wage penalty—in local public administration (− 1.8%). This simple measure already shows that there is higher wage equity between immigrants and the Canadian-born in the public sector.

¹ A complete list of summary statistics is found in Appendix Table 7. Aboriginal and International public administration are small groups that are not discussed in this paper.

Table 1 Selected summary statistics, immigrants and Canadian-born, public and private sectors

	Immigrants		Canadian-born	
	Public sector workers	Private sector workers	Public sector workers	Private sector workers
<i>Real hourly wage (in 2002\$)</i>				
All public/private sector	24.50 (10.054)	18.44 (9.942)	25.25 (9.337)	20.22 (9.998)
Non-administration public sector	23.80 (9.971)	N/A	24.67 (9.198)	N/A
Federal public administration	27.36 (9.772)	N/A	27.58 (9.481)	N/A
Provincial public administration	26.19 (10.088)	N/A	26.16 (9.649)	N/A
Local public administration	26.86 (10.010)	N/A	26.38 (9.376)	N/A
Aboriginal public administration	27.17 (8.636)	N/A	21.83 (9.564)	N/A
International public administration	18.07 (9.416)	N/A	23.42 (10.254)	N/A
Proportion of total employment across sectors	17.75%	82.25%	28.48%	71.52%
Type of public sector employee				
Public employee, not administration	76.30%	N/A	72.09%	N/A
Federal government public administration	9.82%	N/A	10.66%	N/A
Provincial public administration	7.07%	N/A	7.99%	N/A
Local public administration	6.38%	N/A	9.05%	N/A
Aboriginal public administration	0.06%	N/A	0.19%	N/A
International public administration	0.37%	N/A	0.01%	N/A
Highest level of education				
Grade 8 or less	0.40%	2.67%	0.23%	1.23%
Grades 9–10	0.65%	2.64%	0.85%	4.13%
Grades 11–13, non-graduate	0.55%	1.86%	0.90%	3.55%
Grades 11–13, graduate	7.90%	17.61%	10.33%	21.92%
Some post-secondary education	2.70%	4.13%	3.69%	6.27%
Trades certificate/diploma	4.45%	6.68%	8.14%	15.28%
Community college, CEGEP, etc.	22.23%	20.96%	27.82%	25.39%
University certificate below Bachelor's	3.05%	2.56%	3.62%	2.49%
Bachelor's degree	32.67%	27.75%	30.50%	15.35%
Above Bachelor's degree	25.42%	13.14%	13.93%	4.40%
Age				
	4179 (7.923)	4034 (8.234)	4058 (8.541)	3903 (8.820)
	Immigrants		Canadian-born	
	Public sector workers	Private sector workers	Public sector workers	Private sector workers
Sex				
Male	38.44%	52.25%	36.18%	56.32%

Table 1 (continued)

	Immigrants		Canadian-born	
	Public sector workers	Private sector workers	Public sector workers	Private sector workers
Female	61.56%	47.75%	63.82%	43.68%
Marital status				
Married	68.76%	68.77%	53.67%	44.95%
Common-law	7.04%	6.36%	19.49%	21.58%
Widow or widower	0.55%	0.61%	0.53%	0.48%
Separated	3.23%	2.92%	3.18%	3.35%
Divorced	4.86%	4.20%	4.70%	4.28%
Single, never married	15.56%	17.14%	18.44%	25.36%
Multiple job holder				
Single job holder, not a job changer	92.79%	94.90%	94.07%	95.41%
Single job holder, job changer	0.05%	0.04%	0.03%	0.07%
Multiple job holder	7.17%	5.06%	5.90%	4.52%
Permanent or temporary job status				
Permanent	84.47%	91.37%	87.88%	92.66%
Not permanent, seasonal job	0.62%	1.12%	0.87%	2.47%
Not permanent, temporary, term or contract job	11.55%	5.64%	8.53%	3.73%
Not permanent, temporary, casual job or work done through a temporary help agency*	3.17%	1.83%	2.56%	1.08%
Not permanent, other	0.19%	0.05%	0.16%	0.04%
Work schedule				
Full-time (30+ hours/week)	87.92%	90.14%	88.79%	90.69%
Part-time work (< 30 h/week)	12.08%	9.86%	11.21%	9.31%
Firm size				
< 20 employees	3.50%	20.76%	3.14%	22.45%
20–99 employees	7.04%	18.18%	6.53%	19.49%
100–500 employees	11.43%	16.48%	12.81%	15.40%
>500 employees	78.03%	44.58%	77.53%	42.66%
Tenure at current job (months)	96.17 (86.661)	69.84 (74.874)	129.41 (104.235)	89.71 (92.894)
Union status				
Union member	69.99%	14.17%	75.85%	17.71%
Not a union member but covered by a CA	3.09%	1.54%	3.37%	1.73%
Not a union member and not covered by a CA	26.93%	84.29%	20.78%	80.56%
Weighted sample	60,641,954	280,942,331	330,899,875	830,782,228
Unweighted sample		167,056	648,379	1,379,449

Notes: *Two or three categories were combined to meet Statistics Canada's microdata release guidelines. Standard deviations for continuous variables are in parentheses. Complete summary statistics can be found in Appendix Table 7

Within the public sector, about three-quarters of employment (76.3% of immigrants and 72.1% of the Canadian-born) is held in the non-administration public sector. The proportion of immigrants in federal, provincial and local administration are somewhat

lower than among the Canadian-born. Overall, 17.8% of immigrants are employed in the public sector compared to 28.5% of the Canadian-born, likely at least in part the result of more stringent citizenship or residency requirements for some positions in this sector.

Consistent with Nadeau (2013), immigrants record higher levels of education than the Canadian-born, and this does not differ by sector. In the public sector, 58.1% of immigrants employed hold a minimum of a bachelor's degree, whereas the equivalent fraction for the Canadian-born is 44.4%. The proportional difference between the immigrant and the Canadian-born workers in the private sector is even larger, with the fraction of immigrant workers with a bachelor's degree or higher about twice that of the Canadian-born (40.9% versus 19.8%).

Comparing demographics, immigrant workers in both the public and private sectors are on average about 1 year older than the Canadian-born. Among both immigrants and the Canadian-born, females make up more than 60% of public sector employee population, leaving the private sector to be the relatively male-dominated sector. Some 68.8% of immigrants in both the public and private sectors are married, while the equivalent proportions among the Canadian-born are 53.7% and 45.0%, respectively.

Regarding job characteristics, public sector employees in both groups are slightly *more* likely to hold more than one job, *less* likely to hold permanent jobs, and *more* likely to work part-time hours, all of which seem to be contrary to public perception. Job tenure, on the other hand, is higher among public sector workers (especially the Canadian-born), suggesting higher job security in the public sector. Not surprisingly, both immigrant and Canadian-born public sector workers are employed in larger firms (i.e. greater than 500 employees) compared to private sector workers, and about three-quarters of public sector workers are union members or covered by a collective agreement (73.1% for immigrants, 79.2% for Canadian-born). By comparison, less than one-fifth of private sector workers are union members or covered by a collective agreement (15.7% for immigrants, 19.4% for Canadian-born). This may indicate that both immigrants and Canadian-born workers in the public sector have a stronger voice to bargain for higher wages. Lammam et al. (2016) made an observation similar to this.

With regard to the immigrant-specific variables (not shown here), most immigrants originate from non-traditional source countries. Still, while traditional source regions of North America, the Caribbean and Europe are only about 32% of the individuals in the sample, they do account for almost 40% of public sector employment.² A significant proportion of immigrants obtained their highest level of education in Canada, with public sector workers recording a relatively higher fraction (58.2%) than their private sector counterparts (43.8%). Also, 95.2% of immigrants employed in the public sector have been in Canada for 5 or more years, compared to 89.1% in the private sector.

² These figures are obtained by multiplying the proportion of immigrants in each sector from these regions by the corresponding weights at the bottom of Appendix Table 7. All other figures in this section can be found in Appendix Table 7.

Methodology

The outcome variable is log of real hourly wage (in 2002 dollars) which is derived from the hourly earning variable “HRLYEARN” by adjusting it with the monthly consumer price index (CPI) from Statistics Canada. In what follows there are two main comparisons to be made, each with similar models. In all cases included are controls for the highest level of education along with other demographic control variables (sex, marital status, age and its square, family type), job characteristics (multiple job holder status, job tenure, full-time/part-time), union status, survey period (month, year), geography (province, urban status) and occupation. At times, an industry variable will also be introduced to the models (as will be explained below).

The first comparison is the real wage premium for the Canadian-born compared to immigrants by sector of the economy: the private sector, and the four definitions of the public sector listed above. Formally, the human capital model in this case is:

$$\ln(w_i)^s = x_i^s \beta_1^s + CA_i^s \beta_2^s + \varepsilon_i^s \quad (1)$$

where s indicates the private sector or any of the four public sector definitions; x_i is the vector of individual and job characteristics for individual i ; β_1 is a vector of coefficients representing the rates of return to each of the elements in this vector; CA_i is a dummy variable that is equal to one if the individual is Canadian-born and zero otherwise, β_2 the real wage premium for the Canadian-born, and ε_i is the usual error term which is assumed to have a mean of zero and a constant (and finite) variance.

With this model, it can be seen how the Canadian-born wage premium differs between sectors. But as Mueller (2019) highlights in his analysis, the key limitation of this model is that it does not allow one to ascertain the explained (or composition effect) and unexplained (or wage structure effect) portions of the total wage differentials. It is thus uncertain what part of the total wage differential is attributable to differences in labour market characteristics (i.e. individual worker characteristics) between immigrants and the Canadian-born, and what is attributable to the different rates of labour market returns in each group (i.e. how these characteristics are rewarded). The OLS method of estimation assumes that both groups have the same rates of return to worker attributes. This could be a serious limitation in the present case since, for example, the rates of return to education and experience for immigrants may be lower than for the Canadian-born if their foreign credentials are not recognised in Canada.

To address this challenge, this work applies the ubiquitous decomposition technique developed by Blinder (1973) and Oaxaca (1973) to evaluate the average immigrant wage difference across the public and private sectors with the caveat that “... the wage setting model is assumed to be linear and separable in observable and unobservable characteristics” (Firpo et al., 2009:37). The equation is given as:

$$\overline{\ln(w_{CA})}^s - \overline{\ln(w_{IM})}^s = \sum b_{CA}^s (\bar{x}_{CA}^s - \bar{x}_{IM}^s) + \sum (b_{CA}^s - b_{IM}^s) \bar{x}_{IM}^s \quad (2)$$

where again s represents one of the five sectors; the subscripts IM and CA respectively are indicators for the groups of immigrants and the Canadian-born; $\overline{\ln(w_j)^s}$ is the sample mean natural log of real hourly wage for the respective group ($j = \text{IM, CA}$); $\bar{x}_j^s$ is a row vector of the sample mean values of the individual characteristics in each group j in sector s ; and b_j^s is a vector of coefficients of the returns to these characteristics for each group j in sector s .

The explained component of the total wage differential is the first term on the right-hand-side of Eq. (2). This component represents the part of the wage gap due to differences in the average worker characteristics in either group. The second term on the right-hand-side of the equation is the unexplained component. This is the part of the wage gap due to differences in the wage structure (i.e. how worker characteristics are rewarded) in each group sector and is often regarded as a measure of discrimination. In Eq. (2), the immigrant group mean characteristics and the Canadian-born coefficient estimates are used as the reference points. Switching the subscripts above would result in Canadian-born mean characteristics and the immigrant coefficient estimates used as the reference points. The two estimates together provide upper and lower bounds on the Canadian-born wage premium that are explained and unexplained. However, the pay structure of either group may not represent the appropriate counterfactual of the other group (Fortin et al., 2011), so the estimates are weighted by the relative size of each group as proposed by Cotton (1998). This gives estimates between these upper and lower bounds. These are the estimates that are presented and discussed below.³

The second comparison is the public sector wage premium within the immigrant and Canadian-born groups by public sector definition relative to the private sector. The models are essentially the same as those above with some of the subscripts and superscripts juggled to reflect this different emphasis. Now, Eq. (1) becomes:

$$\ln(w_i)^z = x_i^z \beta_1^z + \text{NA}_i^z \delta_1^z + \text{FED}_i^z \delta_2^z + \text{PROV}_i^z \delta_3^z + \text{LOCAL}_i^z \delta_4^z + \varepsilon_i^z \quad (1a)$$

where z denotes whether worker i is an immigrant (IM) or Canadian-born (CA), $\ln(w_i)$ is the natural log of real hourly wages, x_i is a vector of characteristics for individual i , NA_i , FED_i , PROV_i and LOCAL_i are dummy variables representing the unique sector of employment of individual i (i.e. non-administration, federal, provincial and local public administration, with the private sector the omitted category); β_1 is a vector of coefficients representing the rates of return to each of the elements of x_i ; δ_1 – δ_4 are the public sector wage premiums (or penalties if negative) relative to the private sector; and ε_i is the usual error term.

Manipulating Eq. (2) in a similar way results in:

$$\overline{\ln(w_G)^z} - \overline{\ln(w_P)^z} = \sum \beta_G^z (\bar{x}_G^z - \bar{x}_P^z) + \sum (\beta_G^z - \beta_P^z) \bar{x}_P^z \quad (2a)$$

where again z represents whether the individual is an immigrant (IM) or Canadian-born (CA),

³ Other results are available upon request.

the subscripts G and P respectively indicate if an employee is employed in the public sector (by each of the four unique definitions) or the private sector; $\overline{\ln(w_j)^z}$ is the sample mean natural log of real hourly wage in the respective sector ($j = G, P$) for each group z ; $\bar{x}_j$ is a row vector of the sample mean values of the determinants in each respective sector; and β_j is a vector of coefficients representing returns to the various characteristics in each sector.

Empirical Results

In this section, the wage gaps are estimated in two ways in accordance with the models presented above. In the first instance, the immigrant wage gap within each sector is estimated, both private and public, and within each of the subsectors of the public sector. In the second instance, it is the public sector wage premium within both the immigrant and Canadian-born populations that will be estimated.

The Immigrant Wage Gap Within Sectors

OLS Results

Five separate OLS regressions are estimated using Eq. (1): The first for the private sector, followed by one for each of the four public sector definitions. These are all included in Table 2 to facilitate comparisons.⁴

From the results presented, the premium for being a Canadian-born worker is largest within the private sector, with the Canadian-born earning 11.9% more than immigrants with similar characteristics in the same sector. This is higher than the equivalent gap in any of the four public sector definitions.⁵

Within the public sector, the estimates of the immigrant wage gap within the non-administration public sector show a 6.3% premium, almost identical to the 6.6% premium in the federal public administration sector. The provincial public administration records the largest immigrant wage gap of 7.8%. On the other hand, at 4.5%, the immigrant wage gap is smallest for the local public administration sector. In fact, the value of the observed premium for the Canadian-born in the local public administration sector is less than half of the observed equivalent in the private sector. This larger observed immigrant wage gap in the private sector relative to the public sector is consistent with the literature (Nadeau, 2013; Zheng, 2017).

⁴ Full regression results are available from the authors upon request. It should be noted the estimated coefficients on the Canadian-born dummy variable “CA” is an approximate measure of how much more the Canadian-born earn on average compared to immigrants with similar characteristics (i.e., the estimated immigrant wage gap). The “true” percentage marginal effects (ME) can be calculated as $ME = 100 \cdot (e^\gamma - 1)$ where γ is the estimated coefficient. In the case of the first estimate in Table 2, this is calculated as $ME = 100 \cdot (e^{0.119} - 1) = 12.64$ or about a 12.6% wage premium for the Canadian-born. The closer the coefficient estimates are to zero, the more accurate they will be.

⁵ The premiums for the aggregated public sector (i.e., all public sector employees in both non-administration and administration jobs) were also estimated here and below. The estimates were almost identical to those for the non-administration public sector in all cases. Not surprising given that this group accounts for over 70% of all public sector employment for both immigrants and Canadian-born (see Table 1). These results are available upon request.

Table 2 Canadian-born real wage premiums within sectors

	CA born		R ²	Sample size
Private sector	0.119 <i>0.0000</i>	***	0.5090	3,786,573
Non-admin public sector	0.063 <i>0.0001</i>	***	0.4537	1,132,690
Federal public admin	0.066 <i>0.0001</i>	***	0.4027	166,515
Provincial public admin	0.078 <i>0.0002</i>	***	0.4426	127,095
Local public admin	0.045 <i>0.0002</i>	***	0.4500	115,355

Notes: Separate regressions were run for each respective sector, each with the following controls: highest level of education, age and its square, sex, marital status, economic family type, province, urban status, survey period (year and month), multiple job holder, full/part-time job status, job type, firm size, job tenure, union status, occupation and industry (in the case of the first two estimates). The coefficient estimates for aboriginal and international public administration sectors are excluded here since there are few observations and the estimates are not reliable, but they are included as controls in all public sector regressions. Standard errors are italicised beneath each estimate. All standard errors are robust. *, ** and *** denote statistical significance at the 10%, 5% and 1% levels, respectively

Oaxaca-Blinder Decomposition Results

The OLS approach evaluates the wage differentials under the assumption that the wage specification parameters are homogenous across sectors. This however does not always hold, as evidenced in the decomposition results below. Using Eq. (2) to estimate the decomposed immigrant wage gaps, there are significant variations in the rewards for comparable workers across the respective sectors.

As seen in Table 3, the within-sectors pattern of the total immigrant wage gap results is consistent with the patterns from the OLS results above; namely that the overall gap is largest in the private sector at 10.5%. This smaller overall observed gap within the levels of the public sector may be attributable to sectoral differences in the acquisition of Canadian credentials in the public sector. For example, the proportion of immigrants employed in the public sector who attained their highest level of education in Canada and have resided in the country for more years (suggesting more Canadian-acquired work experience) is relatively higher than those employed in the private sector (see Appendix Table 7).⁶ As a result of this Canadian work experience and education, public sector immigrant workers may face lower penalties for their credentials relative to their private sector counterparts.

Additionally, these results show that in the private sector where the immigrant wage gap is largest, the unexplained effect component (i.e. the wage structure effect) accounts for about 80% of the total wage differential (i.e., 0.085/0.105) while the remaining 20% of the total wage gap is attributed to the explained effect (i.e. the composition effect). This larger proportion of the unexplained effect in the private

⁶ An extension of this work might address differences in the wage gap by period of landing.

sector is consistent with Nadeau's (2013) finding. A similar proportional split within the non-administration level of the public sector is also observed; this could be due to positive unobservable factors of the Canadian-born such as talent, motivation and work effort or may be due to negative unobservable factors affecting immigrants such as discrimination or lack of foreign credential recognition.

Within the local public administration sector, however, the explained effect component (i.e. the composition effect) explains a great deal of the total immigrant wage gap observed. This means that the relatively lower wages observed among immigrants within the local public administration can be explained away by the differences in endowment between the two groups of workers. On the contrary, the unexplained effect component (i.e., the wage structure effect) within the local public administration favours immigrants, indicating that immigrants in that sector are paid more than the Canadian-born in way that cannot be explained. The negative coefficient values here could reflect this and could be attributed to unobservable factors that immigrants possess in greater quantities (e.g. work effort, talent).

The Sectoral Wage Gap Within Immigrant Groups

Here we perform an exercise similar to that above by using the same methodologies but addressing the differences in the public sector wage premium within the immigrant and Canadian-born groups.

OLS Results

Using Eq. (1a) estimates of the public sector wage premium among the immigrants and the Canadian-born are presented in Tables 4 and 5, respectively.⁷ In the first case, the sample is limited to immigrants and then the models are built up in a stepwise fashion as more control variables are introduced in each successive iteration. In the second case, the exercise is repeated, but for the Canadian-born. One can observe (in column 1 of each of the tables) that the public-sector wage premium (i.e., the public-private wage gap) is relatively higher among the immigrant group when the two tables are compared. This is consistent with Zheng's (2017) finding. The public-sector premium for each respective group however tends to decline as additional control variables are added to the model.

The education variable across each respective level of the public sector has a relatively large effect on the observed premium among both the Canadian-born and the immigrant groups. When education is controlled for (column 2), the public-sector premium declines in all cases by about five to ten percentage points, reflecting the high levels of education possessed by public employees. Adding in the next group of variables in columns 3 increases the public-sector premium slightly for those in the non-administration sector but decreases the premium in the three public administration groups.

⁷ Here the analysis is limited to addressing the disaggregated public sector. Results which include the aggregated public sector are almost identical to those for the non-administration public sector, reflecting the large number of public sector workers in this category (see Table 1). These results are available upon request.

Table 3 Oaxaca-Blinder decomposition of the immigrant wage gap within sectors

	Total difference		Explained effect		Unexplained effect	
Private sector ($W=0.1712$)	0.105	***	0.020	***	0.085	***
	<i>0.0000</i>		<i>0.0000</i>		<i>0.0000</i>	
Non-admin public sector ($W=0.1147$)	0.050	***	0.012	***	0.039	***
	<i>0.0001</i>		<i>0.0001</i>		<i>0.0001</i>	
Federal public admin ($W=0.0968$)	0.016	***	− 0.025	***	0.041	***
	<i>0.0002</i>		<i>0.0002</i>		<i>0.0002</i>	
Provincial public admin ($W=0.0887$)	0.009	***	− 0.055	***	0.064	***
	<i>0.0002</i>		<i>0.0003</i>		<i>0.0003</i>	
Local public admin ($W=0.0795$)	0.011	***	0.057	***	− 0.046	***
	<i>0.0002</i>		<i>0.0001</i>		<i>0.0002</i>	

Notes: See notes to Table 2. W is the immigrant weight in each sector

When job tenure and union status are entered (columns 4 and 5) the premium decreases further, reflecting more job tenure and higher rates of unionisation among public employees. In the case of unionisation, the effect seems to be small—only 1 to 2.5 percentage points—even though the public perception is often that public sector unions are powerful and able to extract large wage concessions. This union premium is higher in all cases for the Canadian-born.

Adding in the occupational variables (column 6) further reduces the premium by a large amount in each case—from about 4.6 percentage points in the case of Canadian-born local public administration employees, to about 15 percentage points in the cases of immigrants who are employed in federal and provincial administration. This largely reflects the different occupational composition between the private and public sectors (see Appendix Table 7).

Column 7 is the fully loaded model and adds in the industry variables. Here the coefficient estimate on the non-administration public sector decreases slightly. For this definition of the public sector, employment also occurs in the private sector⁸ and as a result, controls for industry can be added and the coefficient estimates easily interpreted (i.e. an 8.8% wage premium for immigrants and a 5.1% premium for the Canadian-born). These premiums in the non-administration public sector may be arguably considered as the “pure public-sector premium” as public workers in this sector are employed in observationally equivalent industries as private sector workers.

But note that these same premiums increase dramatically in the cases of the three levels of public administration. There is a different interpretation for these results. Since public administration forms its own industry, there are no private sector workers

⁸ For example, about 35% of immigrants in the public sector are employed in the health care and social assistance industry, whereas about 8% of immigrants in the private sector are employed in this industry. But given that the private sector employs almost five times as many immigrants than the public sector, there are more immigrants employed in this industry privately than publicly. These numbers can be calculated using the figures and associated weights in Appendix Table 7.

Table 4 Public sector real wage premiums, immigrants

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Non-admin public sector	0.292 <i>0.0001</i>	0.233 <i>0.0001</i>	0.241 <i>0.0001</i>	0.213 <i>0.0001</i>	0.202 <i>0.0001</i>	0.089 <i>0.0001</i>	0.088 <i>0.0001</i>
Federal public admin	0.447 <i>0.0002</i>	0.351 <i>0.0002</i>	0.295 <i>0.0001</i>	0.267 <i>0.0001</i>	0.254 <i>0.0002</i>	0.106 <i>0.0001</i>	0.242 <i>0.0004</i>
Provincial public admin	0.396 <i>0.0002</i>	0.309 <i>0.0002</i>	0.273 <i>0.0002</i>	0.249 <i>0.0002</i>	0.238 <i>0.0002</i>	0.088 <i>0.0002</i>	0.224 <i>0.0004</i>
Local public admin	0.424 <i>0.0002</i>	0.379 <i>0.0002</i>	0.304 <i>0.0002</i>	0.269 <i>0.0002</i>	0.258 <i>0.0002</i>	0.152 <i>0.0002</i>	0.284 <i>0.0004</i>
Education variable	N	Y	Y	Y	Y	Y	Y
Demographic variables	N	N	Y	Y	Y	Y	Y
Province, urban status, year and month	N	N	Y	Y	Y	Y	Y
Multiple job holder, part-time job and job type	N	N	Y	Y	Y	Y	Y
Firm size	N	N	Y	Y	Y	Y	Y
Job tenure	N	N	N	Y	Y	Y	Y
Union status	N	N	N	N	Y	Y	Y
Occupation	N	N	N	N	N	Y	Y
Industry (agriculture, forestry, etc. omitted)	N	N	N	N	N	N	Y
R ²	0.0682	0.1501	0.2748	0.3035	0.3037	0.5185	0.5348

Notes: Separate regressions were run for each respective column. The sample size for each regression is 815,435. The demographic variables include age, age squared, sex, marital status and economic family type. All results are statistically significant at 1% significance level. Robust standard errors are in italics

employed at any of the three levels of public administration (see Appendix Table 7). What is shown here is the public wage premium relative to private sector workers in agriculture, forestry, fishing and hunting industry. But these wage premium estimates are dependent on the choice of omitted industry. Here, a low-wage industry is omitted, resulting in positive public sector premiums. When a high-wage industry is omitted, these premiums become penalties in all cases.⁹

Among the Canadian-born group in administration, the public wage premium is highest at the federal public administration level (18.8%), followed by the local public administration (17.9%) and subsequently the provincial public administration level (15.3%). This order of the highest premium within levels of public administration among the Canadian-born group is consistent with the general finding of Mallett and Wong (2008). Among the immigrant group however, local public administration sector workers enjoy the highest wage premium (28.4%), followed by the federal public administration (24.2%) and then the provincial public administration (22.4%).

⁹ When a high wage industry (mining, quarrying, and oil and gas extraction) is omitted, these numbers are scaled down by about 37 (30) percent points for immigrants (the Canadian-born). See Ansah (2020) for full results.

Table 5 Public sector real wage premiums, Canadian-born

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Non-admin public sector	0.237 0.0000	0.137 0.0000	0.144 0.0000	0.121 0.0000	0.096 0.0000	0.041 0.0000	0.051 0.0000
Federal public admin	0.358 0.0001	0.257 0.0001	0.205 0.0001	0.188 0.0001	0.164 0.0001	0.102 0.0001	0.188 0.0001
Provincial public admin	0.299 0.0001	0.202 0.0001	0.165 0.0001	0.146 0.0001	0.122 0.0001	0.069 0.0001	0.153 0.0001
Local public admin	0.308 0.0001	0.260 0.0001	0.181 0.0001	0.158 0.0001	0.136 0.0001	0.090 0.0001	0.179 0.0001
Education variable	N	Y	Y	Y	Y	Y	Y
Demographic variables	N	N	Y	Y	Y	Y	Y
Province, urban status, year and month	N	N	Y	Y	Y	Y	Y
Multiple job holder, part-time job and job type	N	N	Y	Y	Y	Y	Y
Firm size	N	N	Y	Y	Y	Y	Y
Job tenure	N	N	N	Y	Y	Y	Y
Union status	N	N	N	N	Y	Y	Y
Occupation	N	N	N	N	N	Y	Y
Industry (agriculture, forestry, etc. omitted)	N	N	N	N	N	N	Y
R ²	0.0709	0.1717	0.3424	0.3597	0.3611	0.5002	0.5231

Notes: The sample size for each respective regression is 4,517,643. Refer to other notes under Table 4

Oaxaca-Blinder Decompositions

Table 6 presents a decomposition of the public-private sector wage differential at the mean for immigrants and Canadian-born, respectively, using Eq. (2a). For all levels of the public sector, the explained effect (i.e. the composition effect) is the more dominant component of the total average wage gap. This larger effect of the explained component corroborates most literature which suggests that the public sector tends to employ workers with larger income-generating endowments such as education (Lammam et al., 2016; Tiagi, 2010). Zheng (2017), by contrast, concludes that the public-private sector wage gap observed among the immigrant group is attributable to the dominant component of the unexplained effect (i.e. the wage structure effect).

In terms of the unexplained effect, or the public-sector premium, the patterns of the OLS results above are repeated here for both groups; namely that the premium for immigrants is higher than that for the Canadian-born and that the premium is higher for those in public administration compared to those in non-administration positions.

Conclusion

Using the 12 monthly cycles of the Labour Force Survey from each of the years 2006 through to 2018, this paper evaluates the wage differences between immigrants and the Canadian-born, both within and between the various levels of the public sector and the

Table 6 Oaxaca-Blinder decomposition of the public-private sector wage gap, immigrants and Canadian-born

	Total difference		Explained effect		Unexplained effect	
Non-admin public sector						
Immigrants ($W=0.1669$)	0.292	***	0.193	***	0.099	***
	0.0001		0.0001		0.0001	
Canadian-born ($W=0.2422$)	0.237	***	0.191	***	0.047	***
	0.0000		0.0001		0.0001	
Federal public admin						
Immigrants ($W=0.0243$)	0.447	***	0.335	***	0.112	***
	0.0002		0.0002		0.0002	
Canadian-born ($W=0.0457$)	0.358	***	0.265	***	0.093	***
	0.0001		0.0001		0.0001	
Provincial public admin						
Immigrants ($W=0.0171$)	0.396	***	0.301	***	0.095	***
	0.0002		0.0002		0.0002	
Canadian-born ($W=0.0356$)	0.299	***	0.235	***	0.064	***
	0.0001		0.0001		0.0001	
Local public admin						
Immigrants ($W=0.0139$)	0.424	***	0.232	***	0.192	***
	0.0002		0.0005		0.0005	
Canadian-born ($W=0.0327$)	0.308	***	0.228	***	0.079	***
	0.0001		0.0002		0.0002	

Notes: The following controls were used in each regression: highest level of education, age and its square, sex, marital status, economic family type, province, urban status, survey period, multiple job holder, full/part-time job status, job type, firm size, job tenure, union status, occupation and industry (in the case of non-admin only). *, ** and *** denote statistical significance at the 10%, 5% and 1% levels, respectively. W is the respective public sector weight. Figures in parentheses are standard errors. All standard errors are robust

private sector. The empirical results show that the observed trend of the wage gap in the non-administration public sector differs from the results of the various levels of public administration. This may suggest that previous empirical studies that do not disaggregate the public sector may likely reflect public sector results that are strongly skewed towards the non-administration public sector (i.e. the largest subsector within the public sector) and may not entirely be applicable to the distinct levels of public administration.

The empirical analysis further shows that—after controlling for worker and job characteristics—the average immigrant wage gaps in the respective sub-levels of the public sector are significantly smaller than that of the private sector.

Within the private sector where the immigrant wage gap is largest, the decomposition results show that most of the total gap is due to the unexplained effect. This signals

that most of the gap is not due to differences in worker endowments but rather unobserved factors that cause the wage structure in the private sector to reward immigrant workers less than the Canadian-born. These results reinforce the assertion that the immigrant wage gap may be attributed to factors such as discrimination and the problem of foreign credential recognition that has been highlighted in the literature. Other unobserved factors that may cause the unexplained effect to be larger in the private sector could be talent, motivation, work effort and prestige associated with a job.

Results also show the public wage premium is generally larger among the immigrant group. The higher relative public-sector wage premium among immigrants does not imply that the absolute value of real hourly wage received by immigrants in the public sector is larger than that received by the Canadian-born in the same sector. Rather, it can be seen as a reinforcement to the observation that immigrants in the private sector suffer a higher immigrant wage penalty than those in the public sector, as evidenced in the wider margin of wage differences between immigrants in the public sector and immigrants in the private sector.

Also, the explained effect accounts for most of the observed total public-private sector wage differences among both immigrants and Canadian-born. These decomposition results support the empirical finding that that “individuals who join the public sector tend to be positively selected on observables” (Tiagi, 2010:471).

To address the immigrant wage gap that is attributable to discrimination against immigrant workers, particularly in the private sector, government might consider expanding efforts at implementing the purpose of the Employment Equity Act (S.C. 1995, c.44) as last amended on December 12, 2017. The Act seeks to, *inter alia*, “correct the conditions of disadvantage in employment experienced by. . . members of visible minority [groups] . . .” (Government of Canada, 2020:1). This would include a significant number in the immigrant community.

Additionally, in recent times there is increasing pressure for governments to constrain public-sector wages to balance budgets, it is necessary that such policies are implemented with caution. Given that most of the public-sector premium observed is attributable to the positive value of the component/explained effect, cutting wages at those levels could lead to the loss of such skilled workers to the private sector, to other jurisdiction, or even the emigration of public employees. The result could be a qualitative decline in the public sector and the commensurate loss of productivity.

While this paper adds to literature on immigrant-Canadian-born wage comparisons, there are some limitations of this study. First, the LFS neither contains ethnicity nor language skills variables, both of which can confound immigrant wage outcomes. Also, the public administration industry is exclusive to the public sector and so it has an inherent ability to introduce biases when being compared to the private sector. Future studies on this subject can probe further to measure the marginal effect of discrimination in determining the immigrant wage gap using ethnicity as a proxy, while controlling for immigrant-specific variables to observe how the wage gaps change under those conditions.

Appendix

Table 7 Summary statistics, immigrants and Canadian-born, public and private sectors (standard deviations for continuous variables are in parenthesis)

	Immigrants		Canadian-born	
	Public sector workers	Private sector workers	Public sector Workers	Private sector workers
Real hourly wage (in 2002\$)				
All public/private sector	24.50 (10.054)	18.44 (9.942)	25.25 (9.337)	20.22 (9.998)
Non-administration public sector	23.80 (9.971)	N/A	24.67 (9.198)	N/A
Federal public administration	27.36 (9.772)	N/A	27.58 (9.481)	N/A
Provincial public administration	26.19 (10.088)	N/A	26.16 (9.649)	N/A
Local public administration	26.86 (10.010)	N/A	26.38 (9.376)	N/A
Aboriginal public administration	27.17 (8.636)	N/A	21.83 (9.564)	N/A
International public administration	18.07 (9.416)	N/A	23.42 (10.254)	N/A
Proportion of total employment across sectors	17.75%	82.25%	28.48%	71.52%
Type of public sector employee				
Public employee, not administration	76.30%	N/A	72.09%	N/A
Federal government public administration	9.82%	N/A	10.66%	N/A
Provincial public administration	7.07%	N/A	7.99%	N/A
Local public administration	6.38%	N/A	9.05%	N/A
Aboriginal public administration	0.06%	N/A	0.19%	N/A
International public administration	0.37%	N/A	0.01%	N/A
Highest level of education				
Grade 8 or less	0.40%	2.67%	0.23%	1.23%
Grade 9–10	0.65%	2.64%	0.85%	4.13%
Grade 11–13, non-graduate	0.55%	1.86%	0.90%	3.55%
Grade 11–13, graduate	7.90%	17.61%	10.33%	21.92%
Some post-secondary education	2.70%	4.13%	3.69%	6.27%
Trades certificate/diploma	4.45%	6.68%	8.14%	15.28%
Community college, CEGEP, etc.	22.23%	20.96%	27.82%	25.39%
University certificate below Bachelor's	3.05%	2.56%	3.62%	2.49%
Bachelor's degree	32.67%	27.75%	30.50%	15.35%
Above Bachelor's degree	25.42%	13.14%	13.93%	4.40%
Age	41.79 (7.923)	40.34 (8.234)	40.58 (8.541)	39.03 (8.820)
Sex				
Male	38.44%	52.25%	36.18%	56.32%
Female	61.56%	47.75%	63.82%	43.68%
Marital Status				
Married	68.76%	68.77%	53.67%	44.95%

Table 7 (continued)

Common-law	7.04%	6.36%	19.49%	21.58%
Widow or widower	0.55%	0.61%	0.53%	0.48%
Separated	3.23%	2.92%	3.18%	3.35%
Divorced	4.86%	4.20%	4.70%	4.28%
Single, never married	15.56%	17.14%	18.44%	25.36%
Economic family type				
Unattached individual	10.75%	11.42%	14.48%	17.98%
Husband-wife, dual earner couple, no children or none under 25	14.18%	14.02%	19.72%	20.28%
Husband-wife, dual earner couple, youngest child 0 to 17	40.95%	36.00%	38.94%	31.39%
Husband-wife, dual earner couple, youngest child 18 to 24	7.01%	6.27%	6.54%	5.66%
Husband-wife, single earner couple, husband employed, no children or none under 25	1.48%	2.40%	1.30%	2.47%
Husband-wife, single earner couple, husband employed, youngest child 0 to 17	5.25%	8.58%	2.12%	4.39%
Husband-wife, single earner couple, husband employed, youngest child 18 to 24	0.57%	0.86%	0.41%	0.66%
Husband-wife, single earner couple, wife employed, no children or none under 25	1.62%	1.45%	2.11%	1.88%
Husband-wife, single earner couple, wife employed, youngest child 0 to 17	2.80%	2.18%	1.63%	1.18%
Husband-wife, single earner couple, wife employed, youngest child 18 to 24	0.77%	0.66%	0.58%	0.43%
Husband-wife, non-earner couple, no children or none under 25	0.79%	1.00%	0.74%	1.11%
Husband-wife, non-earner couple, youngest child 0 to 17	0.04%	0.10%	0.01%	0.02%
Husband-wife, non-earner couple, youngest child 18 to 24	0.06%	0.13%	0.05%	0.06%
Single-parent family, parent employed, youngest child 0 to 17	5.46%	4.36%	5.71%	5.11%
Single-parent family, parent employed, youngest child 18 to 24	1.72%	1.43%	1.59%	1.38%
Single-parent family, parent not employed, youngest child 0 to 17	0.08%	0.16%	0.04%	0.09%
Single-parent family, parent not employed, youngest child 18 to 24	0.04%	0.09%	0.04%	0.07%
Other families	6.43%	8.89%	3.99%	5.85%
Province				
Newfoundland	0.27%	0.09%	2.24%	1.68%
Prince Edward Island	0.11%	0.07%	0.63%	0.43%
Nova Scotia	0.95%	0.46%	3.50%	3.13%
New Brunswick	0.59%	0.33%	2.88%	2.58%
Quebec	16.82%	14.53%	26.61%	26.37%
Ontario	47.76%	53.39%	34.23%	35.36%

Table 7 (continued)

Manitoba	4.24%	2.98%	4.54%	3.19%
Saskatchewan	2.07%	1.29%	4.09%	3.02%
Alberta	11.81%	11.34%	10.30%	12.76%
British Columbia	15.39%	15.52%	10.98%	11.48%
Survey year				
2006	6.15%	6.96%	7.52%	7.84%
2007	6.34%	7.06%	7.69%	7.86%
2008	6.57%	7.03%	7.72%	7.90%
2009	6.60%	6.77%	7.69%	7.67%
2010	6.96%	6.95%	7.78%	7.73%
2011	7.64%	7.25%	7.65%	7.78%
2012	7.66%	7.54%	7.63%	7.78%
2013	7.84%	7.77%	7.74%	7.66%
2014	8.23%	7.90%	7.64%	7.66%
2015	8.58%	8.09%	7.74%	7.60%
2016	8.88%	8.67%	7.63%	7.48%
2017	8.92%	8.81%	7.77%	7.57%
2018	9.63%	9.21%	7.80%	7.47%
Multiple job holder				
Single job holder, not a job changer	92.79%	94.90%	94.07%	95.41%
Single job holder, job changer	0.05%	0.04%	0.03%	0.07%
Multiple job holder	7.17%	5.06%	5.90%	4.52%
Permanent or temporary job status				
Permanent	84.47%	91.37%	87.88%	92.66%
Not permanent, seasonal job	0.62%	1.12%	0.87%	2.47%
Not permanent, temporary, term or contract job	11.55%	5.64%	8.53%	3.73%
Not permanent, temporary, casual job or* work done through a temporary help agency	3.17%	1.83%	2.56%	1.08%
Not permanent, other	0.19%	0.05%	0.16%	0.04%
Work schedule				
Full-time (30+ hours/week)	87.92%	90.14%	88.79%	90.69%
Part-time work (< 30 h/week)	12.08%	9.86%	11.21%	9.31%
Firm size				
< 20 employees	3.50%	20.76%	3.14%	22.45%
20–99 employees	7.04%	18.18%	6.53%	19.49%
100–500 employees	11.43%	16.48%	12.81%	15.40%
> 500 employees	78.03%	44.58%	77.53%	42.66%
Tenure at current job (months)	96.17	69.84	129.41	89.71
	(86.661)	(74.874)	(104.235)	(92.894)
Union status				
Union member	69.99%	14.17%	75.85%	17.71%
Not a union member but covered by a CA	3.09%	1.54%	3.37%	1.73%
Not a union member and not covered by a CA	26.93%	84.29%	20.78%	80.56%
Occupation				

Table 7 (continued)

Legislators and senior management	0.23%	0.29%	0.41%	0.44%
Specialised middle manager occupations (1)	0.93%	2.02%	1.22%	2.47%
Specialised middle manager occupations (2)	0.51%	0.94%	0.44%	0.75%
Specialised middle manager occupations (3)	0.45%	0.09%	0.52%	0.13%
Specialised middle manager occupations (4)	1.65%	0.18%	2.58%	0.34%
Specialised middle manager occupations (5)	0.07%	0.04%	0.12%	0.12%
Middle management occupations in wholesale and retail trade and customer service	0.13%	2.18%	0.23%	3.22%
Middle management occupations in trades, transportation, production and utilities (1 and *2)	0.27%	0.67%	0.37%	1.25%
Middle management occupations in trades, transportation, production and utilities (3)	0.13%	0.58%	0.18%	0.73%
Professional occupations in business and finance	4.01%	4.99%	3.34%	4.08%
Administrative and financial supervisors and administrative occupations	6.35%	4.28%	7.70%	5.55%
Finance, insurance and related business administrative occupations	0.41%	1.20%	0.51%	1.35%
Office support occupations	6.65%	4.91%	6.51%	5.01%
Distribution, tracking and scheduling co-ordination occupations	2.17%	2.26%	2.38%	2.15%
Professional occupations in natural and applied sciences	7.20%	8.79%	3.53%	4.42%
Technical occupations related to natural and applied sciences	3.52%	4.38%	3.57%	4.12%
Professional occupations in nursing	9.38%	0.48%	7.22%	0.41%
Professional occupations in health (except nursing)	2.86%	0.58%	2.23%	0.57%
Technical occupations in health	5.01%	1.36%	4.16%	1.44%
Assisting occupations in support of health services	5.70%	2.29%	2.96%	1.42%
Professional occupations in education services	15.43%	0.51%	18.55%	0.55%
Professional occupations in law and social, community and government services	5.23%	1.42%	6.08%	1.80%
Paraprofessional occupations in law and social, community and government services	2.22%	2.21%	2.52%	2.19%
Occupations in front-line public protection services	1.57%	0.01%	4.01%	0.03%
Care providers in education, legal, and public protection support occupations	3.45%	1.13%	3.75%	0.42%
Professional occupations in art and culture	0.58%	0.39%	0.67%	0.64%
Technical occupations in art and culture	0.52%	1.08%	0.94%	1.39%
Retail sales supervisors and specialised sales occupations	0.13%	3.20%	0.19%	4.53%
Service supervisors and specialised service occupations	0.93%	4.75%	0.96%	3.46%
Sales representatives and sales persons - wholesale and retail trade	0.21%	3.97%	0.27%	5.33%
Service representatives and other customer and personal services occupations	1.76%	5.75%	1.56%	5.39%
Sales support occupations	0.28%	3.04%	0.28%	2.46%

Table 7 (continued)

Service support and other occupations, nec	5.09%	5.48%	3.28%	2.94%
Industrial, electrical and construction trades	0.52%	4.44%	1.37%	7.27%
Maintenance and equipment operation trades	0.76%	2.60%	1.15%	5.03%
Other installers, repairers and services and material handlers	0.29%	2.15%	0.27%	2.00%
Transport and heavy equipment operation and related maintenance occupations	2.38%	2.82%	2.00%	4.33%
Trades helpers, construction labourers and related occupations	0.31%	0.81%	0.65%	1.02%
Supervisors and technical occupations in natural resources, agriculture and related production	0.03%	0.23%	0.08%	1.15%
Workers in natural resources, agriculture and related production	0.02%	0.44%	0.06%	0.78%
Harvesting, landscaping and natural resources labourers	0.10%	0.29%	0.25%	0.52%
Processing, manufacturing and utilities supervisors and central control operations	0.52%	1.17%	0.89%	1.56%
Processing and manufacturing machine operators and related production workers	0.01%	4.69%	0.02%	2.64%
Assemblers in manufacturing	0.04%	2.62%	0.01%	1.52%
Labourers in processing, manufacturing and utilities	0.01%	2.29%	0.02%	1.08%
Industry				
Agriculture, forestry, fishing and hunting	0.07%	0.72%	0.28%	1.38%
Mining, quarrying and oil and gas extraction	0.03%	1.16%	0.08%	3.12%
Utilities	2.62%	0.14%	3.68%	0.29%
Construction	0.26%	5.32%	0.57%	9.65%
Manufacturing	0.15%	20.03%	0.08%	16.30%
Wholesale trade	0.02%	4.79%	0.03%	5.79%
Retail trade	0.41%	11.21%	0.61%	13.57%
Transportation and warehousing	6.03%	4.96%	4.61%	5.42%
Information and cultural industries	0.85%	3.05%	0.92%	3.24%
Finance and insurance	1.23%	7.94%	1.23%	6.74%
Real estate and rental and leasing	0.43%	1.66%	0.27%	1.55%
Professional, scientific and technical services & *Management of companies and enterprises	0.25%	10.43%	0.25%	8.35%
Administrative and support, waste management and remediation services	0.23%	5.64%	0.26%	4.33%
Educational services	27.39%	0.85%	29.62%	0.78%
Health care and social assistance	34.82%	8.26%	27.83%	7.90%
Arts, entertainment and recreation	1.28%	0.93%	1.64%	1.49%
Accommodation and food services	0.21%	8.20%	0.14%	5.54%
Other services (except public administration)	0.03%	4.71%	0.01%	4.56%
Public administration	23.70%	N/A	27.91%	N/A
Immigrant-specific characteristics				
Region of Birth				
North America (Excl. Canada)	4.60%	2.44%	N/A	N/A

Table 7 (continued)

Central America	2.20%	3.04%	N/A	N/A
Caribbean and Bermuda	7.41%	5.54%	N/A	N/A
South America	4.41%	5.23%	N/A	N/A
Western Europe	5.43%	3.45%	N/A	N/A
Eastern Europe	8.00%	7.71%	N/A	N/A
Northern Europe	8.64%	5.22%	N/A	N/A
Southern Europe	5.82%	5.97%	N/A	N/A
Western Africa	2.14%	1.39%	N/A	N/A
Eastern Africa	3.49%	2.39%	N/A	N/A
Northern Africa	4.19%	2.98%	N/A	N/A
Central Africa	0.99%	0.62%	N/A	N/A
Southern Africa	0.76%	0.62%	N/A	N/A
West Central Asia and Middle East	5.31%	6.29%	N/A	N/A
Eastern Asia	10.85%	13.86%	N/A	N/A
South-East Asia	13.55%	15.84%	N/A	N/A
Southern Asia	10.84%	16.24%	N/A	N/A
Oceania	1.14%	0.97%	N/A	N/A
Europe (not specified), Africa (not specified), Asia (not specified)	0.21%	0.20%	N/A	N/A
Country/region obtained highest level of education				
Canada	58.19%	43.82%	N/A	N/A
North America (excl. Canada)	3.51%	2.59%	N/A	N/A
Central America	0.58%	0.93%	N/A	N/A
Caribbean and Bermuda	1.13%	1.21%	N/A	N/A
South America	1.49%	2.24%	N/A	N/A
Western Europe	3.47%	3.26%	N/A	N/A
Eastern Europe	3.36%	4.34%	N/A	N/A
Northern Europe	2.53%	2.96%	N/A	N/A
Southern Europe	0.93%	1.29%	N/A	N/A
Western Africa	0.96%	0.65%	N/A	N/A
Eastern Africa	0.41%	0.66%	N/A	N/A
Northern Africa	3.20%	2.34%	N/A	N/A
Central Africa	0.27%	0.31%	N/A	N/A
Southern Africa	0.15%	0.32%	N/A	N/A
West Central Asia and Middle East	2.13%	3.31%	N/A	N/A
Eastern Asia	3.13%	6.49%	N/A	N/A
South-East Asia	7.79%	11.39%	N/A	N/A
Southern Asia	6.10%	11.34%	N/A	N/A
Oceania	0.67%	0.55%	N/A	N/A
Years since became landed immigrant				
Landed for a year or less	0.28%	0.66%	N/A	N/A
Landed for less than 5 years (but more than 1 year)	4.54%	10.24%	N/A	N/A

Table 7 (continued)

Landed for 5–10 years	17.79%	22.19%	N/A	N/A
Landed for more than 10 years	77.39%	66.90%	N/A	N/A
Weighted sample	60,641,954			
280,942,331				
330,899,875				
830,782,228		Unweighted Sample	167,056	648,379
1,379,449	3,138,194			

Notes: *Two or three categories were combined to meet Statistics Canada's microdata release guidelines. Standard deviations for continuous variables are in parenthesis

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Declaration

Conflict of Interest/Competing Interests The authors declare no competing interests.

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